Lecture 11: Evolution, Emotion, and Reason: Emotions, Part I (full transcript)

Source: https://oyc.yale.edu/psychology/psyc-110/lecture-11
License: Creative Commons Attribution-NonCommercial-ShareAlike 3.0 (Open Yale Courses)

Overview: This class is an introduction to the evolutionary analysis of human emotions, how they work, why they exist, and what they communicate. In particular, this lecture discusses three interesting case studies, that of happiness (e.g., smiling), fear and the emotions we feel towards our relatives. Finally, this lecture ends with a brief discussion of babies' emotional responses to their caregivers.

Chapter 1. The Different Functions of Emotions [00:00:00]

Professor Paul Bloom: On Monday we — I presented an introduction to evolutionary psychology, the looking at psychology from an evolutionary perspective, and trying to make a case and give some examples of how it can help illuminate and illustrate certain aspects of how the mind works. One of the advantages of an evolutionary perspective on the mind is that it forces us to look scientifically at what we would otherwise take for granted. There are a lot of aspects of how we are and what we are and what we do that seem so natural to us. They come so instinctively and easily it's difficult, and sort of unnatural, to step back and explore them scientifically but if we're going to be scientists and look at the mind from a scientific perspective we have to get a sort of distance from ourselves and ask questions that other people would not normally think to ask. And the clearest case of this arises with the emotions. And as a starting point there's a lovely quote from the psychologist and philosopher William James that I want to begin with. So, he writes:

To the psychologist alone can such questions occur as: Why do we smile when pleased and not scowl? Why are we unable to talk to a crowd as we talk to a single friend? Why does a particular maiden turn our wits upside down? The common man — [None of you are the common man.] The common man can only say, "Of course we smile. Of course our heart palpitates at the sight of the crowd. Of course we love the maiden. And so probably does each animal feel about the particular things it tends to do in the presence of certain objects. To the lion it is the lioness which is made to be loved; to the bear the she-bear. To the broody hen, the notion would probably seem monstrous that there should be a creature in the world to whom a nestful of eggs was not utterly fascinating and precious and never to be too-much-sat-upon object which it is to her.

Now, there's a few things to note about this passage. First, it's incredibly sexist. It assumes not just merely in reflexive use of phrases. It assumes that — William James assumes he's talking to males, male humans who sometimes take the perspective of male bears. And so, it assumes a male audience. You wouldn't normally — You wouldn't actually ever write this way. A second point is it's beautifully written and you're not — ;also, not allowed to write that way anymore either. It's poetic and lyrical and if — William James characteristically writes that way. I think he writes so much better than his brother, Henry James, an obscure novelist. [laughter] Finally though, the point that he makes is a terrific one, which is yes, all of these things seem natural to us but the reason why they seem natural is not because they are in some sense necessary or logical truths. Rather, they emerge from contingent aspects of our biological nature.

And so we need to step back. We actually — We need to step back and ask questions like — and these are questions we're going to ask — Why does poop smell bad? Avoid the temptation to say, "Well, poop smells bad because it's so stinky." The stinkiness of poop is not an irreducible fact about the universe. Rather, the stinkiness of poop is a fact about human psychology. To a dung beetle poop smells just fine. Why does chocolate taste good? Well, chocolate — The good tastiness of chocolate isn't some necessary fact about the world. It's a fact about our minds that doesn't hold true for many other creatures. And so, we have to step back and ask why to us do we find chocolate appealing?

Why do we love our children? Don't say they're lovable. Many of them are not [laughter] and, as William James points out, every animal, most animals, many animals love their children. They think their children are precious and wonderful. Why? Why do we get angry when people hit us? Suppose somebody walked up to you and slapped you in the face? You'd be afraid. You'd be angry. Would you get sleepy, feel nostalgic, suddenly desire some cold soup? [laughter] No. Those are stupid alternatives. Of course if somebody slapped us you would — we would get angry or afraid. Why? Why do we feel good when someone does us a favor? Why don't we feel angry? Why don't we feel fearful? What we're going to do throughout this course is step back and ask these questions. We're going to ask questions nobody would have otherwise thought to ask, where the common man wouldn't address, and this is, of course, standard in all sciences.

The first step to insight is to ask questions like why do things fall down and not up? And I imagine the first person who articulated the question aloud probably met with the response saying, "What a stupid question. Of course things fall down." Well, yes, of course things fall down, but why? Why is our flesh warm? Why does water turn solid when it gets cold? These are natural facts about the universe, but the naturalness needs to be explained and not merely assumed. In this class we're going to explore, throughout the course, what seems natural to us and try to make sense of it. And to that end we have to ask questions that you wouldn't normally ask. We've already done this to some extent with domains such as visual perception, memory, language and rationality, but now we're going to move to the case where it's maybe even somewhat more difficult to do this. Now, we're going to start dealing with the emotions. We're going to talk about the emotions, why they exist, what they're there for, and how they work.

I want to start off with the wrong theory of the emotions. And the wrong theory of the emotions is beautifully illustrated in the television and movie series Star Trek. In this alternative fantasy world, there are characters, Mr. Spock in the originalStar Trek, Data in one of the spin-offs, who are described as competent, capable, in fact in many ways, super competent and super capable people. But they're described as not having emotions. Spock is described as not having emotions because he's half Vulcan, from a planet where they lack emotions. Data is an android who is said to lack an emotion chip.

This lack of emotions on this — on a TV series does not hurt them much. They're able to fully function. And in fact, in a TV series emotions are often seen as a detriment. You do better off without them. And there are many people in sort of common sense who might think "Gee, if only I could just use my rationality, think reasonably and rationally and not let my emotions guide my behavior I'd be much better off." It turns out that this is a notion of how to think about the emotions that is deeply wrong. And in fact, makes no sense at all.

Using the example of Star Trek, Steven Pinker, in his book How the Mind Works, nicely illustrates the problem here. He writes, "Spock must have been driven by some motives or goals. Something must have led him to explore strange new worlds, to seek out new civilizations and to boldly go where no man had gone before." Presumably, it was intellectual curiosity that set him to drive and solve problems. It was solidarity with his allies that led him to be such a competent and brave officer. What would he have done if attacked by a predator or an invading Klingon? Did he do a handstand, solve the four-color map theorem? Presumably, a part of his brain quickly mobilized his faculties to scope out how to flee and how to take steps to avoid a vulnerable predicament in the future. That is, he had fear. Spock did not walk around naked around the ship. Presumably, he felt modesty. He got out of bed. Presumably, he had some ambitions and drive. He engaged in conversations. Presumably, he had some sociable interests.

Chapter 2. Phineas Gage and the Loss of Emotional Capacity [00:09:06]

Without emotions to drive us we would do nothing at all. And you could illustrate this scientifically. Creatures like Spock and Data don't exist in the real world but there are unusual and unfortunate cases where people lose, to some extent or another, their emotions. And you could look at these people and see what happens to them. The classic case, the most famous case, is that of a man called Phineas Gage. Phineas Gage is the classic Intro Psych example – an extremely poor guy, poor schmuck.

In 1848 — He was a construction foreman. In 1848 he was working at a site with explosives and iron rods. And due to an explosion, an iron rod passed through his head like so. Imagine that rod shooting upwards. It went under his eye and popped out the top of his head. It landed about one hundred feet away covered with blood and brains. The rod itself weighed thirteen pounds. Amazingly, Gage was not killed. In fact, he was knocked unconscious only for a short period and then he got up and his friends surrounded him and asked, "Are you okay?" And they — And then they took him to the hospital. On the way to the hospital, they stopped by a tavern and he had a little pint of cider to drink, sat down and talked to people. And then he had an infection, had to have surgery. But when it was all said and done he wasn't blind, he wasn't deaf, didn't lose language, didn't become aphasic, no paralysis, no retardation. In some sense, what happened was much worse. He lost his character.

Here's a description at the time of what Gage was like. And this is from Damasio's excellent book Descartes' Error:

He used to be a really responsible guy, a family man, very reliable, very trustworthy. But after the accident he was fitful, irreverent, indulging at times in the grossest profanity, manifesting but little deference for his fellows, impatient of restraint or advice, a child in his intellectual capacities and manifestations. He had the animal pleasures of a strong man. His foul language is so debased that women are advised not to stay long in his presence.

And he couldn't hold a job. He lost his family, couldn't hold a job. He ended up in the circus. He was in the circus going around the country with his big iron rod telling everybody the story as they surrounded him and clapped. There are other cases like Phineas Gage, cases where people have had damage to that same part of the brain, parts of the frontal cortex. And what they've lost is they basically lost a good part of their emotions. And what this means is they don't really care that much about things. They can't prioritize.

Damasio tells a case of one of his patients who was under the pseudonym here of Elliot. And Elliot had a tumor in his frontal lobe. And the tumor had to be removed and with it came a lot of Elliot's frontal lobe. And again, as a result of this, Elliot was not struck blind or deaf or retarded, and he didn't become the sort of profane character that Phineas Gage became, but he lost the ability to prioritize. He lost the ability to set goals. Damasio describes him here:

At his job at an activity he would read and fully understand the significance of the material [He works in an office.] but the problem was he was likely, all of a sudden, to turn from the task he had initiated to doing something else and spending an entire day doing that. He might spend an entire afternoon deliberating on which principle of categorization he should apply to files. Should it be the date or the size of the document, pertinence to the case or another?

He couldn't set his goals. He couldn't — He ended up not being able to keep a job, not being able to deal with people. And these are not men who have lost their emotions. There is no case around where you could have your emotions entirely blotted out. But they lost a large part of their emotional capacity and as a result, their rationality failed.

Emotions set goals and establish priorities. And without them you wouldn't do anything, you couldn't do anything. Your desire to come to class to study, to go out with friends, to read a book, to raise a family, to be — to do anything are priorities set by your emotions. Life would be impossible without those emotions. And so, there's certain themes we're going to explore here. The first is this, that emotions are basically mechanisms that set goals and priorities and we're going to talk a lot about — in this class and the next class about universals. We're also going to talk about culture. It turns out that cultures, different cultures, including differences between America and Japan and the American South and the American North, have somewhat different emotional triggers and emotional baselines to respond to. But at the same time, as Darwin well knew, emotions have universal roots that are shared across all humans and across many animals.

So, the agenda for this class and the next class is going to go like this. First, I want to talk a little bit about facial expressions, which are ways in which we communicate our emotions – not the only way, but an important way – and look, in particular, at the case of smiling because it's kind of interesting. Then I want to look at one case study of a nonsocial emotion, that of fear. I want to then deal with feelings towards our kin, people we're genetically related to, and then — and this will take us to the next class, feelings towards non kin.

Chapter 3. Facial Expressions and Smiles in Particular [00:15:45]

So first, faces. And as an introduction to faces I have a brief film clip from Paul Ekman, who is one of the world's great scholars in the study of facial expressions. [clip playing]

In Ekman's work, he presents us with instructions on how to make different faces and identify faces. Ekman actually has a sort of more practical career along with his scientific career. He trains police and secret service members to try to figure out cues to honesty and dishonesty. There's a very interesting New Yorker profile on him by Malcolm Gladwell a few years ago, something you might be interested in. But let's do one of his faces.

Please lower your brows and draw them together. That means even those who aren't making eye contact with me now. Tense your lower and upper eyelids. Don't pop out contact lenses but just tense them. Stare. Your eyes can bulge somewhat. [laughter] Okay. Now, the last part is important. Press your lips together with the corners straight or down. That's good. You got it. [laughter] Okay. Just because you are not making eye contact with me doesn't mean I can't see you. Okay. [laughter] Well, what you're looking like presumably is this [referring to a slide]. And what face is that? What emotion does that correspond to? Anger.

There's all sorts of databases of different faces from around. This guy — I don't know who he is but he seems to be on a lot of these things [laughter] but the thing is you don't need to rely on him. You don't need to rely on Western faces. Even if you go on line there's, by now, a lot of databases from faces from all sorts of genders and national origins. This is from a Japanese women facial expressions. And there are some subtle and very interesting differences across countries and across people, but there's also deep universals. You don't have to work very hard to figure out what these different facial expressions mean.

I want to give one more face example because I want to focus on this a little bit. This one's a little bit easier. Raise the corners of your lips back and up, please. [laughter] Raise your cheeks. Raise your lower eyelids if you can. [laughter] They're smiling. You're smiling. You can stop [laughter] smiling. Yale is actually really big on smiling. We have two of the world's experts on smiling. This is Angus Trumble, the curator at the British Art Gallery who wrote this wonderful book, A Brief History of the Smile looking at the smile in art. And this is my colleague, Marianne LaFrance, who is actually not smiling in that picture but she studies smiling and smiling in adults, smiling in children, smiling across cultures, and the different social uses of smiling. And there are some interesting discoveries people have made about smiles and about smiles and the emotions.

One — Oh. Well, one is that smiles are universal. We know, for instance, that young children smile. This is my son, Zachary, when he was younger, not that weird-looking kid [laughs] next to him. [laughter] Thank God. [laughter] And even blind children, children blind from birth, will smile. They'll smile appropriately, making an important point that smiling is not learned by looking at other people's faces.

Smiling is also not uniquely human. Nonhuman primates smile as well. Smiles are social signals. You might imagine that people smile when they're happy. This is actually not the case. It's not as simple as that. Rather, people smile when they wish to communicate happiness and we know that from several studies. There are some studies of bowlers and the studies are very nice. What they do is they film bowlers. So, the bowlers do their bowling and sometimes they knock down all the pins, which is called a what? A strike. So a strike — and that's good in the bowling world. So, they knock down all the pins but what they don't do, is they don't smile after they knock down the pins. They are being filmed. They don't smile. Then they turn around to their friends and give a big grin.

Other studies have looked at films of people who have just won Olympic gold medals. Now, not surprisingly, people who have won Olympic gold medals are very happy. This is good news to win an Olympic gold medal. But they don't actually stand on the podium grinning. Rather, they stand there with their faces in a normal expression. Then when they stand up and face the crowds, there's a big smile. You can ask yourself whether during sex, an activity where many people enjoy, whether or not

people smile during sex. And you can discover this yourself with [laughter] a partner or a mirror. [laughter]

So, there are other things we know about smiles. There are different types of smiles. There are actually quite a few different types of smiles that are different in interesting ways. This is Paul Ekman again. Which one's a better smile? Who votes for the one on the right? Who votes for the one on the left? There are two different sorts of smiles. The one on the right is a smile of greeting. It's sometimes known as a "Pan Am" smile. Pan Am is a now defunct airline which had at that time — They were — They don't call them stewardesses anymore but they're — the stewardesses would come in and they would smile. That was part of their job. But it was a big, fake smile, the Pan Am smile, a smile to communicate "hello" and — but it's as opposed to a smile where the communication is that of genuine happiness. The difference is around the eyes. It's not the mouth. It's the eyes.

A real happiness smile, what's known as a Duchenne smile, after a neurophysiologist who studied it, involves moving the eyes. What's interesting is about only one out of every ten people can fake a Duchenne smile. So, if you smile at somebody, and you just hate their guts but you want to smile at them, it's — unless you're quite gifted it's difficult to fake a really good, really happy smile.

You could — It's not difficult to study smiles in the real world. You could look at politicians, for instance. Politicians are often in contexts where they have to smile a lot. And what they do is they simply give the Pan Am smile. The mouth moves up, particularly if somebody is attacking their record or ridiculing them, and they'll smile and — but it's not a sincere smile. The eyes don't move.

My favorite example of this was a few years ago when there was a huge battle for the House majority leader. And a guy named — a Republican named John Boehner won this position in quite a heated battle. And they took a picture of the guy — This is not very nice. They took a picture of the guy, Roy Blunt, as he stepped out. And he had lost and this was his expression. [laughter] And he's not really very happy [laughter] as opposed to a smile like this, which is a real smile.

So, you have two sorts of smiles: A real happiness smile [or] a Duchenne smile — called — also known as the Duchenne smile, and then a Pan Am smile, or greeting smile. And you'll use each of those smiles at different points in your day and in your life. It turns out that these different smiles have real psychological validity. They seem to sort of reflect deep differences in your mood and emotions and thoughts. Ten-month-olds, for instance, give different sorts of smiles. When their mother approaches there they give a real happiness smile. Then when a stranger approaches or someone else approaches there they'll tend to give more of a greeting smile.

John Gottman studied married couples. And John Gottman does a lot of work — Well, what he does is he looks at film clips of couples. And by analyzing the film clips he tries to predict will their marriages survive. And one of his cues — There's different cues. Incidentally, sort of side topic: The death knell for a marriage for Gottman — This is his big finding. It's not if they fight a lot. It's not they scream at each other. It's not even if they hate each other. The death knell of a marriage is contempt. And so, if he shows these clips: I walk in, "Honey, I'm home," and my spouse has the look of contempt, it's a bad sign. [laughter] But another clue is the sort of smiles they give when they see each other when they walk into the lab. If it's a true happiness smile, that's actually bodes better for the relationship than a Pan Am, or greeting smile.

Finally, studies have been done of college yearbook photos looking at people thirty years later. And it turns out that there's a correlation, a reliable relationship between how happy somebody is now and back thirty years ago in their yearbook photo — what sort of smile they're giving.

There is some evidence for a third sort of smile. This is known as a coy smile or an appeasement smile. This is sort of a very specialized sort of smile. This is a smile of embarrassment or stress. You give it when you want people to like you, you want to join in; you want to make people feel positive about you. But you're in, sort of, a high-stress situation often with some sort of risk. And what you do is you sort of you turn away. There's no eye contact. You turn away and kind of give this — [demonstrating by tilting his head to the side]

And this actually shows up in other primates. Here's a nice picture. [laughter] So, the rhesus monkey bites her own infant, and the infant gives a scream and then the submissive, coy smile. And it also shows up in human infants. Here's a nice clip of a coy baby smile. I'll walk you through it. The baby is being approached, [laughter] goes like

Chapter 4. Question and Answer on Smiles [00:28:08]

this [with locked-gaze the brows raise and a smile starts], smiles like this [smile widens and head turns up a left], and then the aversion [smile widens and head is further averted]. Yeah. Babies are cute. [laughter] Any questions at this point about smiling? What are your smiling questions? [laughter] Yeah.

Student: Do nonhuman primates' smiles [inaudible]

Professor Paul Bloom: That's a good question. I don't know. There's evidence that the coy smile shows up in non — The question was, "Do nonhuman primates give the same smiles that humans do?" such as a distinction between the Pan Am smile, a greeting smile, versus a genuine smile of happiness? I don't know. I'll find out for you for next class though. That's a good question. Yeah.

Student: How come some people's smiles are better than other people's smiles?

Professor Paul Bloom: How come some people's smiles are better than other people's smiles? The non-interesting psychological answer, some people are better looking and there's more thing — [laughter] but the deeper answer is some people are better able to smile. Some people are better able to use the cues to express true happiness.

There's something else about smiles which is going to come up, which your question raises, I think, which is going to come up in — when we talk about emotional contagion and actually, some issues of morality. Smiles are extremely contagious. So, what I'd like people to do — If you're sitting next to somebody, please turn around and find someone next to you and look at them. Don't do anything. Just look at them. Whoever is being looked at, look back. [laughter] This is not — [laughter] Please arbitrarily decide. Okay. Please arbitrarily decide on the smiler. That will be — No, not at me, at each other, [laughter] and that will be the person — If you are unable to resolve this dispute — yes, you two, please — if you are unable to resolve this dispute, the person to the right of me will be the smiler. So, look at each other expressionless. [laughter] Now, the person who is the mandated smiler, [laughter] on three, please smile. One, two, three. [laughter] Okay. [laughter]

Worst class demo ever [laughter] but if one could imagine more restrained circumstances, it is actually extremely difficult to be facing somebody who's really smiling at you and not smile. This is true, by the way, for virtually every other emotion. The phenomena is known as "emotional contagion," where if you're facing somebody, for instance, and they're — they look at you in a face of absolute rage, it is very difficult to just sit there without your own face molding in accord to their own. And the reasons why this happens and how that works is something we'll talk about later on. So that's — One more question. Yes.

Student: [inaudible]

Professor Paul Bloom: I don't know if that's — The question is, "Is there a difference between smiling with your teeth versus just your lips closed?" There probably is. That's not a main smile difference but my bet is that there probably is a difference. And my bet also is that that sort of distinction, how much teeth you show when you smile, is the sort of thing that would show regional and country by country differences. For instance, there's been research finding that people in England smile different from people in the United States. And I think that those are the sort of contrasts that you would expect to find in cross-cultural differences. Every culture is going to have Pan Am smiles, happiness smiles, coy smiles, but the variation of that sort is something which will vary as a result of how you're raised and the people around you.

Chapter 5. Non-social Emotions: Fear [00:32:22]

I want to deal with a few emotions in this class and next and the first case study of an emotion I want to deal with is the emotion of fear. And I want to deal with fear for different reasons. One reason is it's a basic emotion, it's universal. All humans have it. Many nonhumans, probably most nonhuman, species have it too. And it also brings us back to the lecture on behaviorism where we talked about classical conditioning and different theories of what people are afraid of. It's a nonsocial emotion. What I mean by this is it's possible, of course, to be afraid of a person, but unlike an emotion like gratitude, it's not intrinsically social. You could be afraid of falling off a cliff or something. It has a distinctive facial expression again.

This is a famous picture of Lee Harvey Oswald who was being assassinated by Jack Ruby. And this is the detective's face standing there, a mixture of fear and anger – the face being drawn back in a universal expression that every human everywhere would be able to recognize.

So, the basic question to ask is "What are we afraid of?" And the answer's a little bit interesting. We're afraid of spiders, snakes, heights, storms, large animals, darkness, blood, strangers, humiliation, deep water, and leaving home alone. We are afraid of other things too but those are big things to be afraid of. I'm not even going to ask. If there's somebody who — in this room, who's not afraid of any of those things? You're a tougher person than I am. These are universal fear elicitors. Why? What do they have in common? Why would you be afraid of those things? And the answer is — And why would — why are there so few people afraid of guns, cars, and electrical outlets?

The answer is not particularly surprising. These are things that — something's ticking over there. [referring offstage] These are things that are scary in our ancestral environment. More particularly, these are things that through the course of human evolution have been dangerous to us. And so, we are

afraid of these things [pointing to a slide containing spider, snakes, heights etc.] and not so afraid of these things [pointing to guns, cars, electrical outlets, etc.]; similarly for nonhuman primates. So, chimpanzees are afraid of certain things and they can often develop phobias for certain things, but the phobias they develop, the fears they develop, are things like spiders and snakes.

There was a nice study done in urban Chicago, in the inner city of Chicago. And they asked children raised in the inner city, "What are you most afraid of?" And you might think they would say, "I'm afraid of being shot. I'm afraid of guns. I'm afraid of being killed by somebody or being harmed by somebody. I'm afraid of being run over by a car." The two biggest fears of children in urban Chicago are that [pointing to the slide], snakes and spiders, even though many of these children have probably never seen a snake outside of a zoo in their lives. These are natural fears.

There is some research done by the psychologist Judy DeLoache at University of Virginia where she's studying babies' fears of spiders and snakes, babies obviously who, since their parents are normal, have not yet seen spiders and snakes. There are various ethical reasons why you can't show babies — you can't try to construct phobias in babies of spiders and snakes but the research she's finding using more indirect methods finds, as one would expect, these are what psychologists would call "pre-potent stimuli"; that is, these are things that naturally elicit fear and concern. And that's all I have to say about fear.

Chapter 6. Social Emotions and Altruism [00:36:29]

I want to turn for the rest of this lecture and for next lecture next week to the social emotions. And the social emotions can be broken down into two categories. [a student sneezes] Bless you, bless you. Those emotions you feel towards your kin, towards your genetic relatives, and those emotions that you feel towards the people you're not related to but interact with. And I want to focus particularly on emotions that generate kind or altruistic behavior.

"Altruism" is the biologists' term meaning kindness, generosity, and evolutionary biologists have worked really hard to explain why animals might evolve to be kind. A very old, very wrong view of evolution is that evolution has shaped animals such that they're merely survival machines. If so, then from an evolutionary standpoint any kindness towards an animal — that an animal shows towards another animal — is a mystery. If evolution wired us up simply to survive, then it's a puzzle why animals would relate positively to other animals. But of course, that's not true.

Here's a simple example showing it's not true. Imagine two genes, two sorts of animals each containing their own gene. Gene "A" makes an animal care for its offspring. Gene "B" makes an animal care only for itself. Imagine what will happen in the next generation. Plainly, Gene "A" will win out. It's a very simple case. An animal who has evolved a brain that says, "Take care of your offspring" will do much better from a natural selection point of view from an animal who has evolved a brain that says, "Eat your offspring." The animal that eats its offspring, those genes are a biological dead end. What matters then is not survival, per se. What matters is reproduction. And so, that simple fact is why we would expect animals to care for their children, because children are the means through which genes replicate.

But it gets a little bit richer than that. And this is one of the major revolutions in evolutionary biology over the last half century. Forget about the animal a bit and take another perspective. Take a perspective of the cold virus. People have been sneezing in the front row. Now, you're coughing. Thank you. Why do you sneeze when you get a cold? Here's not a — [more coughing, laughter] Point made. Here is — Here's not a bad answer. You sneeze because you've got all these germs inside you and your body wants to get the germs out, so you sneeze. It's not that it's totally wrong, but it's not bad.

The real answer is a little bit more interesting. Don't look at it from the person's perspective. If you have a cold, try to get away from your own selfish perspective, "I have a cold." Look at it from the perspective of the cold virus. The cold virus has evolved just as much as you evolved. And it's evolved due to survival and reproduction. What the cold virus does is evolve different strategies to cause it to reproduce. And what it does is — one way to reproduce is to occupy other animals and manipulate their bodies so as to expel it. From this point of view then, the reason why you sneeze when you have a cold is that your cold — the cold virus is using your body as a tool to replicate itself. From this person — this perspective, a person is just a germ's way of making other germs. And there's tons of other examples of this.

There's a parasite known as toxoplasmosis that lives in the bodies of rats. But it gets passed on when the rats get eaten by cats. And then it ends up in the cats' feces and then it ends up back in rats. If you are a rat and you have toxoplasmosis, you are perfectly healthy except for one thing. The toxoplasmosis rewires your brain and it makes you less afraid of cats. Now, again, this is not some sort of bizarre quirk of a humorous god. Rather, it's because this is a perfectly — this is the adaptive strategy of the toxoplasmosis virus.

In fact, a real powerful virus would skip the respiratory system altogether, even better than a cold virus. What it would do is it would take over the brain and it will make people want to run around and have sex with other people and kiss them on the mouth. And in fact, there is some evidence that this happens. There's some evidence, for instance, that one of the effects of sexually transmitted diseases like syphilis is it arouses the libido, makes people more sexually engaged, because this is part of the strategy through which these viruses replicate themselves. Imagine a virus, for instance, that captured an animal's brain and then modified the animal's brain such that the animal would run out and bite other animals so as to pass on the virus. And then, of course, you would call that virus "rabies." Along these lines, the evolutionary biologist Richard Dawkins took the general step of suggesting that animals are the vehicles through which genes exploit to reproduce. From this perspective, an animal is just the person's — is just the gene's way of creating another animal.

Well, as psychologists, what benefit does that — does this way of analysis give us? It actually can help us explain altruism. So, which genes are going to survive? Well, the genes that survive are going to be the ones that make the most copies of themselves. Animals are vehicles through which genes reproduce. An animal's merely the gene's way of making another gene. Hence, selfish genes will lead to altruistic animals because, to the extent that evolution operates at the level of the genes, there's no hard and fast distinction between your own body and someone else's body.

And here's an illustration by the biologist Haldane. So, Haldane was once asked, "Would you lay down your life for your brother?" And he responded, "No, but I would gladly give my life for three brothers or five nephews or nine first cousins." Now, he's joking. You don't actually do the math if you're normal.

But what he's capturing is the logic, the ultimate causation of our feelings towards our kin. Our genes have wired us up — our brains up to love our children and love our kin because, in this way, our genes manage to replicate themselves. And in fact, you get his calculations by looking at genetic relatedness.

The genetic relatedness, from an evolutionary standpoint, affects how much you care for other people. From the standpoint of your genes, you dying for the life of three brothers is an excellent compromise because the genes replicate by fifty percent more. If you imagined — ;So, here is his calculations. If you imagined a choice between this one gene that makes the animal choose to die and the other gene that makes an animal choose for its brothers to die, the gene that sacrifices the body it belongs to will make more copies in the future. And there's an interesting irony to this. The selfish gene theory is often seen as sort of a cold-blooded evolutionary analysis, but it provides a scientific basis for real, genuine altruism, for really arguing that, from the standpoint of the genes, there really is no hard and fast difference between yourself and another person.

From this perspective, we can start to answer some interesting questions at least about nonhumans. When a new male lion takes over a pride what he does is kill all the remaining cubs and any lionesses undergo spontaneous abortions. This all might seem very cruel but from a genetic standpoint it makes sense. The other cubs are genetic competition for him. They do not have his genes. Moreover, only once they're out of the way can he reproduce and copulate with the females. The females do their spontaneous abortions because that's a reliable adaptive trick. These cubs are not going to survive once they are born so the female's best strategy is to get rid of them and start anew.

From a psychological point then, animals have evolved to be nice to their kin, particularly their children, and particularly in birds and mammals. Birds and mammals invest in quality and not quantity, as opposed to fish and reptiles. For birds and mammals, we don't have many kids but — so we devote a huge amount of psychological energy to protecting the ones that we have. Moreover, the kids we have are vulnerable for long periods of time and require our resources. So, there's various psychological mechanisms that this gives rise to. One is how parents or how adults in general respond to children. Another one is how children respond to parents. And I'll briefly talk about a few of these phenomena.

[plays a sound of a baby crying] Small animals make distress calls. They chirp, they mew, they bleat or they cry. The governing of a distress call is actually an extremely delicate high-wire act for any young organism from an evolutionary point of view. It has to on the one hand be annoying enough to actually generate help, to get people to help you, to feed you, to pick you up, to take you and put them next to you. On the other hand, it can't be so annoying that the people around you kill you [laughter] and so it's complicated. But, from your point of view, you're wired up to respond to them. That sound is, at very minimum, extremely annoying. And it's more — it's not annoying because of its volume or pitch. It's annoying because your brains are wired up so that that baby cry is going to drive you up the wall.

On the more positive side, babies are cute. I got this [picture] from Google Images, typing in "cute baby," [laughter] getting rid of the porn and [laughter] coming on to that. No, no, Playboy, but anyway it was over that. [laughter] And do not be tempted to say, "Isn't it wonderful that the way nature works is that babies are cute? Otherwise we would have killed them." [laughter] That's not the right story. If — Babies are not — Human babies are not, sort of, metaphysically cute. If Martians came down they

wouldn't say, "Oh, cute baby." Rather, they're cute because of how our brains are wired up. They're cute because there are certain cues that correspond to the way our brains work.

And in fact, this is how it works for all mammals. So, babies have these big, protruding foreheads, an upturned little nose, chubby cheeks and big eyes. Those are the ingredients for cute. Stephen Jay Gould has a wonderful essay where he discussed this, looking at the evolution of Mickey Mouse from the Walt Disney character. Mickey Mouse starts off as an ugly, little rodent. [laughter] Over time he gets cuter and cuter and cuter as the artist converged on more and more baby-like features.

Studies of adults show what's known as a baby-face bias. This is not unique to the United States. The same studies have been done in Asia. You find a baby face in an adult, Leonardo DiCaprio, to be particularly naïve, helpless, kind and warm. And in mock trials, people with baby faces are more likely to be found innocent than people like Ben Affleck, [laughter] who do not have baby faces. [laughter] Now, one question which is going to come up for an entire lecture later on is "who is sexier, the baby faced man or Testosterone Man?" here [pointing to Ben Affleck]. [laughter] And I am going to ask actually for a vote because I'm going to return to this. I do not — I only want the men to vote, please. Who would go for — And forget the fact that he looks sort of unhappy. Who would go for Ben Affleck here? Okay. [laughter] Who would go for Leonardo? Okay. [laughter] Well, the women votes would actually be more complicated. We will discuss when we get the lecture on sex. Your choice will depend on where you are in the menstrual cycle. [laughter]

Now, so far, we're talking about how babies respond to — We're talking about our responses to babies. What about babies' responses to us? Well, there's a very old theory known as the "Cupboard Theory" proposed by the behaviorist B.F. Skinner which argues that babies' attachment to their parents is because the parent provides food, characteristically breast milk, but it could be food from a bottle or whatever. And because of operant conditioning, the baby is driven towards the adult. An alternative theory is that of Bowlby, which is that they're drawn to their mother for comfort and social interaction as well as fear of strangers.

To test this, the psychologist Harlow performed a series of ingenious experiments with nonhuman primates distinguishing between what he called "wire mothers" and "cloth mothers." And you'll see illustrations of this to follow. Wire mothers are mothers that are built that they give food. They have a little nipple attached and you can drink from it and give food to the baby. And that's the baby's source of food. Cloth mothers don't give any food but they give warmth and comfort.

There was a while in the psychology department where one professor was known to be extremely supportive to his students but didn't really provide much warmth. And he was known as the cloth mother. And another one was very productive and everything but provided no love. And she was known as the wire mother. But anyway, I'll show you the movies. [movie playing]

I have to warn you this third and final movie is an example of why this research is not currently done, but it illustrates an important scientific point. Oh. Now him — [movie playing]

I think I'm — [showing pictures of cute baby animals] They're just more Google Images. [laughter] I think I'll — I want to begin next class by wrapping up and explaining the Harlow studies in more detail and what they tell us. And then we'll move towards altruism, towards non kin. I'll see you next week.